Did Liquidity Govern Stablecoin Peg Pass-Through? Evidence from the March 2023 USDC Depeg*

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Abstract

This study asks whether liquidity governed how stablecoin peg stress passed through to the stablecoin price of Bitcoin during the March 2023 USDC depeg. Using 1-minute data on BTC/USD, BTC/USDT, BTC/USDC, USDT/USD and USDC/USD, we decompose each stablecoin's cross-currency basis into its peg deviation and a triangular arbitrage deviation. We then model the arbitrage deviation's response to the peg deviation as a function of lagged illiquidity. During the crisis, the peg deviation accounted for almost all of the basis variance, while the triangular arbitrage deviation decayed within one sampling interval at both 1-minute and 5-minute sampling, against half-lives of 1 to 5 hours for the peg deviations. Liquidity did not measurably govern the pass-through. The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust. The stress that reached the stablecoin price of Bitcoin was the peg deviation itself, not a failure of arbitrage.[1]

1 Introduction

A stablecoin is a cryptocurrency pegged to a fiat currency, at one US dollar for Tether (USDT) and USD Coin (USDC). Exchanges quote Bitcoin against both the dollar and stablecoins, so the stablecoin price of Bitcoin is a dollar price only while the peg holds. Stablecoin pegs do fail, and depegs spill over to other crypto-assets. Perez Riaza and Gnabo [15] estimate that a Tether depeg raises the probability of a jump in BTC/USD almost fivefold within 5 minutes, Foley et al. [8] find that downward Tether depegs lower the returns of major cryptocurrencies, and Bartůněk [3] documents that depegging raises the realized volatility of Bitcoin.

When a peg breaks, triangular arbitrage ties BTC/USDC to BTC/USD and USDC/USD, so BTC/USDC should move away from BTC/USD by the peg deviation. However, arbitrage is costly. Makarov and Schoar [14] document large, recurrent arbitrage opportunities across cryptocurrency exchanges, Reynolds et al. [18] find deviations from triangular arbitrage parity in Bitcoin markets but not among fiat currencies, and Lyons and Viswanath-Natraj [13] estimate that Tether's peg deviations halved once arbitrage with the issuer became easier. If thin order books limit triangular arbitrage, a depeg should pass slowly and incompletely into the stablecoin price of Bitcoin when the peg deviation is largest. This study asks whether liquidity governed how stablecoin peg stress passed into the stablecoin price of Bitcoin.

We address this question using the USDC depeg in March 2023. Silicon Valley Bank, which held a substantial part of the reserves of USDC [15], failed on 10 March 2023 [4]. USDC traded as low as 0.863 dollars on 11 March, before the joint statement of 12 March protected all depositors of the bank [20]. We decompose each stablecoin's cross-currency basis into its peg deviation and a triangular arbitrage deviation. We then model the response of the triangular arbitrage deviation to the peg deviation as a function of lagged illiquidity.

Liquidity did not measurably govern the pass-through of the stablecoin peg. During the crisis, the peg deviation accounted for almost all of the basis variance, and the triangular arbitrage deviation decayed within one sampling interval at both 1-minute and 5-minute sampling, against half-lives of 1 to 5 hours for the peg deviations. Triangular arbitrage therefore remained effective

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[1]Code and replication data are available at *GitHub*.

even when the USDC pairs were illiquid, and the stress on Bitcoin's stablecoin price came from the peg deviation itself. The estimated effect of USDC/USD illiquidity is significant at the 1% level at 1-minute sampling but only at the 10% level at 5-minute sampling.

Section 2 describes the data and the depeg events. Section 3 states the methodology, and Section 4 reports the results. Section 5 concludes with the implications for stablecoin regulation.

2 Data

2.1 Sample and notation

We use 1-minute data from Binance.US for five pairs, BTC/USD, BTC/USDT, BTC/USDC, USDT/USD and USDC/USD, from 1 to 21 March 2023. Each 1-minute bar records the high, low, close, base volume and quote volume of a pair. Write $P_t^{X/Y}$ for the close price of one unit of X in units of Y at time t, and $Q \in \{\text{USDT}, \text{USDC}\}$ for a stablecoin. Deviations are in basis points (bps), so log prices are scaled by $m = 10^4$.

A minute without a trade has zero volume and carries a stale price, which is the non-synchronous trading problem of Lo and MacKinlay [12]. Since a deviation constructed from three closing prices is meaningful only when all three are current, we define F_t as an indicator that all three pairs of the triangle trade at time t, and G_t as an indicator that both t and $t-1$ have $F = 1$,

$$F_t = \mathbf{1}\{v_t^{\text{BTC/USD}} > 0,\ v_t^{\text{BTC}/Q} > 0,\ v_t^{Q/\text{USD}} > 0\}, \qquad G_t = F_t F_{t-1}, \tag{1}$$

where v is the base volume of a pair. This rule is a stricter form of the refresh-time sampling of Barndorff-Nielsen et al. [2], which samples once every asset has traded.

2.2 Depeg event identification

Following Perez Riaza and Gnabo [15], we sample the Q/USD close π_k every 5 minutes. A bar is a breach when $|\pi_k - 1| \geq \tau$, with $\tau = 0.5\%$ in the baseline [15, 17]. An event starts at the first breach bar if the previous bar is at exactly 1 dollar, and otherwise at the last earlier bar at exactly 1 dollar. This adapts the retrospective adjustment of Perez Riaza and Gnabo [15] in their Section 6.1. We merge breaches at most 20 minutes apart and drop events shorter than 60 minutes.

Estimates for one stablecoin use its own window, and estimates on all five pairs use the union of the two windows, which equals the USDT window. Table 1 reports the baseline windows and two alternatives, a ±1.0% threshold and a crossing start at the first breach bar. Minutes before a window form the pre-crisis regime, minutes inside it the crisis regime, and minutes after it the post-crisis regime.

Stablecoin	Threshold and start	Start	End	Hours
USDT	±0.5%, last-peg start (baseline)	10 Mar 21:10	15 Mar 13:10	112.0
USDT	±1.0%, last-peg start	10 Mar 21:10	12 Mar 22:40	49.5
USDT	±0.5%, crossing start	11 Mar 00:55	14 Mar 11:15	82.3
USDC	±0.5%, last-peg start (baseline)	11 Mar 01:20	14 Mar 02:30	73.2
USDC	±1.0%, last-peg start	11 Mar 01:20	13 Mar 18:45	65.4
USDC	±0.5%, crossing start	11 Mar 04:00	13 Mar 18:55	62.9

Table 1: Depeg windows from the 5-minute Q/USD close. Times are UTC in March 2023.

2.3 Liquidity measures

We measure illiquidity with the Kyle estimator of Kyle and Obizhaeva [11], which Brauneis et al. [5] find, together with the Amihud illiquidity ratio, best estimates liquidity levels in cryptocurrency markets. For a pair with 1-minute log return r_t and quote volume V_t,

$$\mathrm{Kyle}_t = m \left(\frac{\frac{1}{5}\sum_{i=0}^{4} r_{t-i}^2}{\sum_{i=0}^{4} V_{t-i}} \right)^{1/3}, \tag{2}$$

so that Kyle_t rises with volatility per unit of volume. The Corwin–Schultz spread estimator of Corwin and Schultz [6] (CS), which, together with that of Abdi and Ranaldo [1], best tracks the time variation of liquidity [5], enters the pass-through regression with the realized volatility (RV). Appendix A defines both.

Table 2 reports that the crisis raised illiquidity most in the USDC pairs. The Kyle illiquidity of USDC/USD rose from 1.23 to 7.23 and that of BTC/USDC from 3.08 to 6.37, compared with a rise from 0.79 to 1.02 for BTC/USD. The quote volume of USDC/USD also rose from 1.13 to 9.76 million dollars per day.

	Kyle			Quote volume per day		
Pair	pre	crisis	post	pre	crisis	post
BTC/USD	0.79	1.02	0.88	141	254	333
BTC/USDT	1.14	1.40	1.73	55.4	113	58.1
BTC/USDC	3.08	6.37	5.33	4.15	5.38	3.71
USDT/USD	0.91	1.48	1.44	3.21	8.99	3.89
USDC/USD	1.23	7.23	3.79	1.13	9.76	0.94

Table 2: Liquidity by pair and regime over the union window. Kyle is the regime mean of Eq. (2), and quote volume is in millions of the quote currency.

3 Methodology

3.1 Basis decomposition

For stablecoin Q, we decompose the cross-currency basis D_t into the peg deviation δ_t and the triangular arbitrage deviation b_t,

$$D_t = m\left(\ln P_t^{\mathrm{BTC}/Q} - \ln P_t^{\mathrm{BTC/USD}}\right), \qquad \delta_t = m \ln P_t^{Q/\mathrm{USD}}, \qquad b_t = -D_t - \delta_t. \tag{3}$$

Triangular arbitrage parity, $P^{\mathrm{BTC/USD}} = P^{\mathrm{BTC}/Q} P^{Q/\mathrm{USD}}$, implies $b_t = 0$ [18]. Full pass-through of the peg therefore means $D_t = -\delta_t$, and b_t is the part of the basis that the peg deviation does not explain. The variance of the basis decomposes into

$$\mathrm{Var}(D) = \mathrm{Var}(\delta) + \mathrm{Var}(b) + 2\,\mathrm{Cov}(\delta, b). \tag{4}$$

3.2 Half-life of deviations

For $x \in \{\delta, b\}$, we estimate a first-order autoregressive model and the half-life HL in minutes [13],

$$x_t = c + \rho x_{t-1} + u_t, \qquad \mathrm{HL} = \frac{\ln 0.5}{\ln \rho}. \tag{5}$$

3.3 Pass-through regression

The pass-through regression is

$$\Delta b_t = \alpha + \beta b_{t-1} + \left(\lambda_0 + \boldsymbol{\lambda}_1^\top \tilde{\mathbf{z}}_{t-1}\right) \delta_{t-1} + \boldsymbol{\theta}^\top \tilde{\mathbf{z}}_{t-1} + \kappa \widetilde{\mathrm{RV}}_{t-1} + \varepsilon_t, \tag{6}$$

where the vector $\mathbf{z}_{t-1}$ holds the lagged CS and Kyle illiquidity of BTC/Q and the Kyle illiquidity of Q/USD, and RV_{t-1} is the lagged RV of BTC/USD. A tilde marks standardization within the estimation sample. We estimate Eq. (6) by least squares with heteroskedasticity and autocorrelation consistent (HAC) standard errors with 60 lags (12 on 5-minute bars).

The coefficient β measures the reversion speed of b, and λ_0 the response of Δb_t to the lagged peg deviation at average liquidity. The vector $\boldsymbol{\lambda}_1$ measures how that response changes with a 1 standard deviation rise in each illiquidity measure, so liquidity governs the pass-through when $\boldsymbol{\lambda}_1 \neq \mathbf{0}$.

3.4 Spillover index and Granger causality

We model the joint dynamics of the five pairs by a vector autoregression (VAR) on 5-minute log returns scaled by m,

$$\mathbf{r}_t = \mathbf{c} + \sum_{l=1}^{p} \Phi_l \mathbf{r}_{t-l} + \boldsymbol{\varepsilon}_t, \qquad \mathrm{E}\left[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t^\top\right] = \Sigma, \tag{7}$$

estimated in each regime of the union window with the lag order $p \leq 6$ selected by the Bayesian information criterion (BIC). The generalized forecast error variance decomposition of Koop et al. [10] and Pesaran and Shin [16] measures the share of the H-step-ahead forecast error variance of pair i due to shocks to pair j,

$$\phi_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(\mathbf{e}_i^\top A_h \Sigma \mathbf{e}_j\right)^2}{\sum_{h=0}^{H-1} \mathbf{e}_i^\top A_h \Sigma A_h^\top \mathbf{e}_i}, \qquad \tilde{\phi}_{ij} = \frac{100\,\phi_{ij}}{\sum_l \phi_{il}}, \tag{8}$$

where A_h are the moving-average coefficient matrices, σ_{jj} is the jth diagonal element of Σ, and $\mathbf{e}_i$ selects pair i. We set $H = 10$ (50 minutes), and the total spillover index of Diebold and Yilmaz [7] is

$$S = \frac{1}{5} \sum_{i \neq j} \tilde{\phi}_{ij}. \tag{9}$$

Pair j Granger-causes pair i [9] when the lags of j help predict i in Eq. (7), which we test with a Wald test that all p coefficients of j in the equation of i are zero. Appendix C states the full formulas and reports all directed tests.

4 Empirical results

4.1 Decomposition of the basis

Figure 1 plots D, δ and b around each crisis window. Before the depeg, all three series stayed close to zero. Once the peg broke, the basis D mirrored the peg deviation δ almost exactly, while the triangular arbitrage deviation b fluctuated around zero with short spikes.

Table 3 reports the decomposition by regime. During the crisis, the correlation of D with $-\delta$ is 0.99 for both stablecoins, and $\mathrm{Var}(\delta)/\mathrm{Var}(D)$ is 1.02 for USDT and 1.00 for USDC, so the peg deviation accounts for almost the entire variance of the basis. Before the crisis, the same correlation is only 0.59 for USDT and 0.04 for USDC because the pegs barely moved and the basis consisted mostly of noise in b. In dollar terms, BTC/USDC and BTC/USDT therefore traded at the

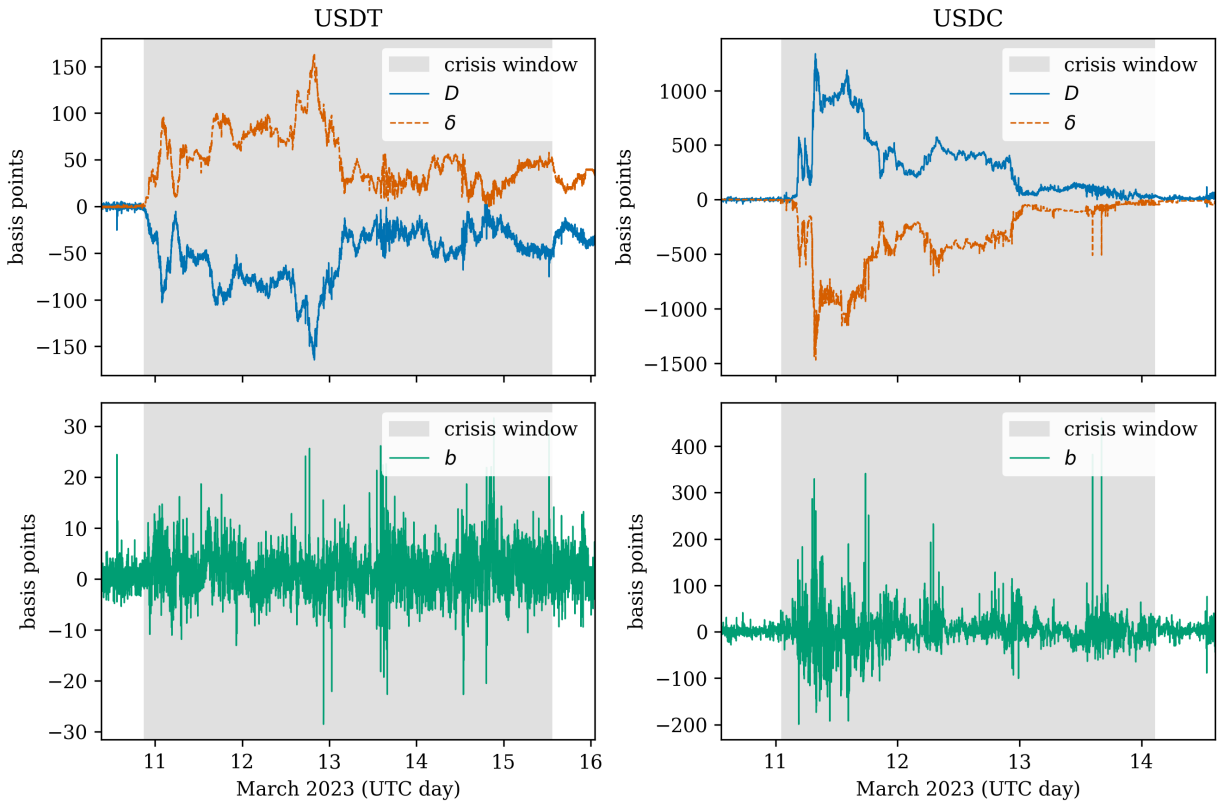


Figure 1: Cross-currency basis D and peg deviation δ (top), and triangular arbitrage deviation b (bottom) in bps, 1-minute close prices from 12 hours before to 12 hours after each crisis window (shaded), UTC.

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
Mean D	−0.11	−53.2	−26.0	0.13	399	7.23
Mean δ	0.12	51.6	25.3	−0.61	−400	−7.17
Var(δ)/Var(D)	0.56	1.02	0.97	0.67	1.00	0.63
Mean b	−0.01	1.57	0.70	0.47	1.72	−0.06
$\lvert b\rvert$ median	1.03	2.50	2.39	2.23	13.4	5.10
$\lvert b\rvert$ 99th percentile	5.37	12.3	10.9	17.3	140	26.5

Table 3: Decomposition of Eq. (3) on synchronous minutes ($F_t = 1$), in bps. Standard deviations and the remaining variance terms are in Table B1.

BTC/USD price throughout the crisis, and the gap in their quoted prices reflected the stablecoin's value rather than a separate Bitcoin price.

The crisis widened the distribution of b but kept its mean below 2 bps, at 1.57 for USDT and 1.72 for USDC, while for USDC the median of $|b|$ rose from 2.23 to 13.4 bps. Triangular arbitrage deviations thus became larger and more frequent but reverted within minutes, as expected from transient trading frictions rather than a sustained mispricing.

4.2 Half-lives

Table 4 reports the half-lives. During the crisis, the half-life of b is 0.82 minutes for USDT and 0.52 minutes for USDC, compared with 158 and 64 minutes for δ. Before the crisis, both half-lives were short, with peg deviation half-lives of 5.7 and 0.9 minutes, since the pegs then only fluctuated around 1 dollar. Arbitrage therefore closed b within a minute even at the height of the stress, while the peg deviation persisted for hours. The persistence of the basis was hence that of the peg deviation.

		pre	crisis	post
USDT	δ	5.72	158	27.8
	b	0.45	0.82	0.64
USDC	δ	0.92	63.9	2.47
	b	0.43	0.52	0.61

Table 4: Half-life in minutes from Eq. (5) on consecutive synchronous minutes ($G_t = 1$).

4.3 Liquidity and pass-through

Table 5 reports the pass-through regression. The coefficient β is between -0.59 and -0.90 and significant at the 1% level in every regime. Reversion was slower during the crisis than before, closing 59% versus 81% of b_{t-1} per minute for USDT and 74% versus 90% for USDC, but remained fast.

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
β	-0.810^{***}	-0.589^{***}	-0.688^{***}	-0.898^{***}	-0.744^{***}	-0.691^{***}
	(-13.77)	(-18.92)	(-23.47)	(-13.42)	(-12.84)	(-7.88)
λ_0	-0.0782^{***}	-0.0098^{***}	-0.0387^{**}	-0.7004^{***}	0.0015	0.0780
	(-2.82)	(-2.88)	(-2.53)	(-7.67)	(0.24)	(1.26)
λ_1, CS of BTC/Q	0.0161	-0.0019	-0.0096^{**}	0.2642^{***}	-0.0006	-0.0079
	(0.61)	(-1.02)	(-2.09)	(2.63)	(-0.13)	(-0.24)
λ_1, Kyle of BTC/Q	0.0124	-0.0045^{*}	0.0205^{**}	0.0735	0.0027	-0.0220
	(0.34)	(-1.65)	(2.20)	(1.08)	(1.06)	(-1.25)
λ_1, Kyle of Q/USD	0.0301^{*}	-0.0009	-0.0060	0.1195^{***}	0.0137^{***}	-0.0317
	(1.89)	(-0.31)	(-1.01)	(7.52)	(2.91)	(-1.06)
Joint p, $\boldsymbol{\lambda}_1 = \mathbf{0}$	0.208	0.263	0.021	<0.001	0.007	0.568

Table 5: Pass-through regression of Eq. (6) on consecutive synchronous minutes, with HAC t statistics in parentheses. Joint p is the p-value of the Wald test of $\boldsymbol{\lambda}_1 = \mathbf{0}$. *, ** and *** denote significance at the 10%, 5% and 1% levels.

For USDT during the crisis, λ_0 is -0.0098 and significant at the 1% level. At the mean premium of 51.6 bps, this implies a move in b of about -0.51 bps per minute. The liquidity terms are jointly insignificant, so the data do not reject an unchanged response (joint $p = 0.263$).

For USDC during the crisis, λ_0 is 0.0015 and insignificant, so at average liquidity the peg deviation did not significantly predict the next change in b. The element of $\boldsymbol{\lambda}_1$ on the Kyle illiquidity of USDC/USD is 0.0137, and the liquidity terms are jointly significant. At the mean discount of -400 bps, a 1 standard deviation rise in this illiquidity implies a move in b of about -5.5 bps per minute. The CS and Kyle illiquidity of BTC/USDC are insignificant, so the effect runs through the USDC/USD pair, in which the peg is set, rather than through the Bitcoin pair. When USDC/USD was illiquid, the peg therefore passed into BTC/USDC with a small delay. The effect is small relative to the discount and not robust to 5-minute sampling (Section 4.5).

The liquidity terms are also jointly significant for USDC before the crisis and for USDT after it.

4.4 Spillovers and lead-lag relations

The BIC selects 6, 2 and 4 lags in the three regimes, and the total spillover index barely moved, at 39.7, 37.5 and 39.4. The crisis did not change overall connectedness but did change its composition. Table 6 reports the share of BTC/USDC forecast error variance due to BTC/USD, which fell from 32.4% before the crisis to 21.4% during it and returned to 31.1% after it. USDC/USD took part of its place, explaining 5.9% of the forecast error variance of BTC/USDC during the crisis, while BTC/USDC explained 10.3% of that of USDC/USD.

The Granger tests point to the same conclusion. BTC/USD led BTC/USDC in every regime and never the reverse, while USDC/USD and BTC/USDC led each other only during the crisis. In the USDT triangle, BTC/USDT led USDT/USD in every regime. Relative to BTC/USDC, price discovery for Bitcoin therefore stayed in BTC/USD throughout, while during the crisis BTC/USDC partly became a market for USDC itself, so that BTC/USDC and USDC/USD returns predicted each other.

	pre	crisis	post
Panel A. Cross variance shares (%)			
Share of BTC/USDC from BTC/USD	32.4	21.4	31.1
Share of BTC/USDC from USDC/USD	0.0	5.9	0.2
Share of USDC/USD from BTC/USDC	0.2	10.3	0.3
Panel B. Granger causality, Wald statistic			
BTC/USD → BTC/USDC	17.79***	10.11***	28.77***
BTC/USDC → BTC/USD	3.51	1.89	7.54
USDC/USD → BTC/USDC	1.37	14.87***	5.27
BTC/USDC → USDC/USD	6.00	114.19***	7.06
BTC/USD → BTC/USDT	16.76**	6.99**	29.54***
BTC/USDT → BTC/USD	10.06	13.57***	3.05
BTC/USDT → USDT/USD	28.45***	15.22***	28.42***
USDT/USD → BTC/USDT	5.15	0.03	4.84

Table 6: Spillovers of 5-minute returns over the union window. Shares are the normalized $\tilde{\phi}_{ij}$ of Eq. (8) with $H = 10$. *, ** and *** denote significance at the 10%, 5% and 1% levels. The full matrices and all Granger tests are in Appendix C.

4.5 Robustness checks

Table 7 reports the key crisis estimates under two variants. R1 re-estimates Eqs. (5) and (6) on 5-minute bars, and R2 enters each liquidity measure alone. At 5-minute sampling, the half-life of b rises to 2.47 minutes for USDT and 2.27 minutes for USDC, still within one sampling interval, and that of δ to 284 and 219 minutes. The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust. Its coefficient stays similar, but at 5-minute sampling it is significant only at the 10% level, and the liquidity terms are jointly insignificant, with a joint p of 0.106. Entered alone, the Kyle illiquidity of USDC/USD remains significant at the 1% level, while neither BTC/USDC measure is significant.

	USDC pass-through			Half-life, USDT / USDC		
Check	λ_1	Joint p	USDT λ_0	b	δ	Share
R0 baseline	0.0137^{***} (2.91)	0.007	-0.0098^{***} (-2.88)	0.82 / 0.52	158 / 63.9	21.4
R1 5-minute bars	0.0143^{*} (1.83)	0.106	-0.0132^{***} (-3.16)	2.47 / 2.27	284 / 219	–
R2 CS of BTC/USDC	0.0005 (0.11)	0.910	-0.0084^{**} (-2.36)	–	–	–
R2 Kyle of BTC/USDC	0.0040 (1.33)	0.185	-0.0091^{**} (-2.56)	–	–	–
R2 Kyle of USDC/USD	0.0149^{***} (2.93)	0.003	-0.0093^{***} (-2.70)	–	–	–

Table 7: Key crisis estimates, with HAC t statistics in parentheses. λ_1 is the element on the Kyle illiquidity of USDC/USD, or on the single measure in R2, and Share is the percentage of BTC/USDC forecast error variance due to BTC/USD. *, ** and *** denote significance at the 10%, 5% and 1% levels.

5 Conclusion

We asked whether liquidity governed how stablecoin peg stress passed through to Bitcoin's stablecoin price during the March 2023 USDC depeg. We decomposed the cross-currency basis of USDT and USDC on 1-minute data into the peg deviation and the triangular arbitrage deviation, estimated their half-lives, and examined whether the triangular arbitrage deviation's response to the peg deviation depended on lagged illiquidity.

During the crisis, the peg deviation accounted for almost all of the variance of the basis, and the triangular arbitrage deviation closed within one sampling interval at both 1-minute and 5-minute sampling, against half-lives of 1 to 5 hours for the peg deviations. Liquidity did not measurably govern the pass-through. The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust. Triangular arbitrage remained effective through the crisis, and the stress that reached the stablecoin price of Bitcoin was the peg deviation itself. Relative to BTC/USDC, price discovery stayed in BTC/USD, while BTC/USDC partly decoupled from it and moved with USDC/USD.

The results also bear on the Guiding and Establishing National Innovation for U.S. Stablecoins (GENIUS) Act of 2025, which brings payment stablecoins under a federal framework and requires one-to-one reserve backing and public disclosure of reserves [19]. Rules that make reserves credible and redemption reliable act on the peg deviation, the component of the basis that carried the stress.

The study covers only one exchange and one episode, so the results may not extend to venues with deeper USDC books or to slower depegs. The liquidity measures are estimated from 1-minute bars rather than observed in the order book. Future work could use order book depth across several exchanges and separate the persistent peg deviation from microstructure noise.

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A Liquidity and volatility measures

Write $\overline{P}_t$ and $\underline{P}_t$ for the high and the low of bar t. The CS estimator over bars $t-1$ and t is

$$\mathrm{CS}_t = m \max\left\{0, \frac{2(e^{\zeta_t}-1)}{1+e^{\zeta_t}}\right\},$$

where

$$\begin{aligned}
\zeta_t &= \frac{\sqrt{2\psi_t}-\sqrt{\psi_t}}{3-2\sqrt{2}} - \sqrt{\frac{\omega_t}{3-2\sqrt{2}}}, \\
\psi_t &= \sum_{i=0}^{1}\left(\ln\frac{\overline{P}_{t-i}}{\underline{P}_{t-i}}\right)^2, \\
\omega_t &= \left(\ln\frac{\max(\overline{P}_t,\overline{P}_{t-1})}{\min(\underline{P}_t,\underline{P}_{t-1})}\right)^2.
\end{aligned}$$

The RV over the last 120 bars is

$$\mathrm{RV}_t = m\left(\sum_{i=0}^{119} r_{t-i}^2\right)^{1/2}.$$

B Additional estimates

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
SD D	2.17	30.4	8.46	4.12	327	10.4
SD δ	1.63	30.7	8.31	3.36	327	8.28
SD b	1.80	4.08	3.78	5.21	38.3	8.83
$\mathrm{Var}(b)/\mathrm{Var}(D)$	0.68	0.02	0.20	1.60	0.01	0.72
$2\mathrm{Cov}(\delta, b)/\mathrm{Var}(D)$	−0.24	−0.04	−0.17	−1.27	−0.01	−0.35
$\lvert b\rvert$ 95th percentile	3.32	8.47	7.72	9.76	81.0	17.3

Table B1: Standard deviations (SD) and remaining decomposition moments on synchronous minutes ($F_t = 1$), in bps.

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
θ, CS of BTC/Q	0.001	−0.011	0.273**	0.041	−1.895	−0.695
	(0.05)	(−0.11)	(2.01)	(0.40)	(−1.01)	(−1.05)
θ, Kyle of BTC/Q	0.025	0.315*	−0.537**	−0.033	0.366	−0.502*
	(0.70)	(1.79)	(−2.07)	(−0.26)	(0.25)	(−1.69)
θ, Kyle of Q/USD	0.016	−0.106	−0.075	−0.220	5.517***	−0.081
	(0.58)	(−0.70)	(−0.44)	(−1.48)	(2.86)	(−0.14)
κ	0.158**	−0.056	0.075	0.298*	0.989	1.442***
	(2.54)	(−0.44)	(0.76)	(1.79)	(0.86)	(2.76)
R^2	0.408	0.290	0.342	0.442	0.373	0.378

Table B2: Control coefficients and fit of Eq. (6), with HAC t statistics in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

C Spillover index and Granger causality

Write the VAR of Eq. (7) in its moving-average representation,

$$\mathbf{r}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} A_h \boldsymbol{\varepsilon}_{t-h}, \qquad A_0 = I, \qquad A_h = \sum_{l=1}^{\min(h,p)} \Phi_l A_{h-l}, \tag{10}$$

where $\boldsymbol{\mu}$ is the mean of $\mathbf{r}_t$. The generalized forecast error variance decomposition of Eq. (8) uses A_h and Σ without orthogonalizing the shocks, so it is invariant to the ordering of the pairs [16]. Since the generalized shares of a row need not sum to 1, each row is normalized, so that $\sum_j \tilde{\phi}_{ij} = 100$ and $\sum_{i,j} \tilde{\phi}_{ij} = 500$. The total spillover index of Eq. (9) is the average share of forecast error variance that comes from the other four pairs [7].

For Granger causality from pair j to pair i, let $\boldsymbol{\gamma}_{ij} = ([\Phi_1]_{ij}, \ldots, [\Phi_p]_{ij})^\top$ collect the p coefficients of j in the equation of i. Under the null hypothesis $H_0 \colon \boldsymbol{\gamma}_{ij} = \mathbf{0}$, the Wald statistic satisfies

$$W_{ij} = \hat{\boldsymbol{\gamma}}_{ij}^\top \left[\widehat{\mathrm{Var}}\left(\hat{\boldsymbol{\gamma}}_{ij}\right)\right]^{-1} \hat{\boldsymbol{\gamma}}_{ij} \xrightarrow{d} \chi^2_p, \tag{11}$$

where $\widehat{\mathrm{Var}}(\hat{\boldsymbol{\gamma}}_{ij})$ is the corresponding block of the estimated covariance of the VAR coefficients.

Row i, from column j	BTC/USD	BTC/USDT	BTC/USDC	USDT/USD	USDC/USD
Pre-crisis					
BTC/USD	34.4	34.0	31.5	0.1	0.0
BTC/USDT	34.0	34.4	31.5	0.1	0.0
BTC/USDC	32.4	32.4	35.1	0.1	0.0
USDT/USD	0.4	0.4	0.5	98.3	0.3
USDC/USD	0.3	0.2	0.2	0.1	99.2
Crisis					
BTC/USD	41.6	40.6	17.4	0.1	0.3
BTC/USDT	40.4	41.6	17.1	0.8	0.2
BTC/USDC	21.4	21.1	51.4	0.2	5.9
USDT/USD	2.8	4.3	1.1	90.5	1.3
USDC/USD	0.5	0.4	10.3	1.3	87.5
Post-crisis					
BTC/USD	35.5	34.6	29.5	0.2	0.1
BTC/USDT	34.6	35.7	29.3	0.2	0.1
BTC/USDC	31.1	30.9	37.6	0.2	0.2
USDT/USD	1.5	1.6	1.1	95.6	0.2
USDC/USD	0.1	0.2	0.3	0.8	98.5

Table C1: Normalized generalized variance shares $\tilde{\phi}_{ij}$ in percent, row pair i from column pair j, 5-minute returns, $H = 10$.

Cause → effect	pre	crisis	post
BTC/USD → BTC/USDT	16.76**	6.99**	29.54***
BTC/USD → BTC/USDC	17.79***	10.11***	28.77***
BTC/USD → USDT/USD	24.22***	22.33***	35.03***
BTC/USD → USDC/USD	9.51	6.38**	5.59
BTC/USDT → BTC/USD	10.06	13.57***	3.05
BTC/USDT → BTC/USDC	9.23	10.92***	2.39
BTC/USDT → USDT/USD	28.45***	15.22***	28.42***
BTC/USDT → USDC/USD	4.74	1.07	8.15*
BTC/USDC → BTC/USD	3.51	1.89	7.54
BTC/USDC → BTC/USDT	3.03	1.77	8.44*
BTC/USDC → USDT/USD	4.06	0.78	3.01
BTC/USDC → USDC/USD	6.00	114.19***	7.06
USDT/USD → BTC/USD	5.21	2.83	8.45*
USDT/USD → BTC/USDT	5.15	0.03	4.84
USDT/USD → BTC/USDC	6.76	9.28***	4.87
USDT/USD → USDC/USD	1.11	5.68*	18.88***
USDC/USD → BTC/USD	2.46	0.93	6.16
USDC/USD → BTC/USDT	2.64	1.30	5.90
USDC/USD → BTC/USDC	1.37	14.87***	5.27
USDC/USD → USDT/USD	8.75	1.98	3.80

Table C2: Granger causality Wald statistics W_{ij}, 5-minute returns, with χ^2 degrees of freedom equal to the lag order (6, 2 and 4). *, ** and *** denote significance at the 10%, 5% and 1% levels.